
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

Anonymous Author(s)

Affiliation

Address

email

Abstract

Forecast accuracy does not by itself establish scientific forecasting ability. We test five conditions that connect a score to a scientific claim: held-out generalization, complete failure accounting, matched disclosure, structural identifiability, and cross-system transfer. We apply these tests to k -pendulum ($k \in \{1, 2, 3\}$) and Lorenz-63 forecasts from language models, a time-series foundation model, learned dynamics, sparse identification, and classical integrators. Each test changes a substantive conclusion. Learned-model error rises from about 0.26 rad on training trajectories to 0.97–1.14 rad on held-out trajectories, where linearized dynamics and nearest-neighbor lookup perform better. Retaining failed forecasts lowers the reported performance of models with incomplete coverage. An unmatched comparison says that hiding physical constants helps; a paired intervention instead finds a +0.069 rad penalty. Two exact scale symmetries prove that the absolute constants are not identifiable from angle trajectories. Increasing the Neural ODE training set improves error from 1.090 to 0.861 rad but does not reach known-equation integration. The pendulum ordering also fails to transfer to Lorenz-63, where SINDy recovers the polynomial dynamics to numerical precision. On video-tracked hardware the roles invert: the RK4 model with published constants is itself imperfect, and its lead over simple baselines is gone within a second. These results show that a correct leaderboard can still support incorrect scientific conclusions. We release the prompts, outputs, checkpoints, seeds, and compute records needed to repeat the audit.

1 Introduction

Scientific claims require more than low forecast error. A result may depend on training trajectories, dropped failures, unmatched inputs, an unidentifiable target, or one favorable system. We study these failure modes directly. Given state $\mathbf{s}_t$, the task is to predict $\mathbf{s}_{t+T}$. The k -link pendulum is integrable at $k=1$ and chaotic for $k \geq 2$, has high-precision ground truth, and permits controlled changes to physical metadata. Lorenz-63 provides a second system with a different observation model and hypothesis class; a hardware check (§5.5) scores the reference model itself as the imperfect one.

Our contribution is a claim audit for dynamical-system forecasts. It tests held-out generalization, counts every failed forecast, records the information available to each method, changes disclosure on identical trajectories, checks identifiability before scoring parameter recovery, varies the training budget, and tests whether conclusions transfer to a second system. Table 1 states the conclusion before and after each test. The central result is empirical and direct: all five tests alter or reject a conclusion supported by the unaudited comparison. These are conclusions the field actually draws: zero-shot forecasters are routinely scored without failure accounting or matched controls [Gruver et al., 2023, Zhang and Gilpin, 2025], and our setting mirrors that practice.

Table 1: Conclusions before and after the claim audit. Each row changes one part of the evaluation while preserving the relevant model outputs or comparison set.

Test	Unaudited conclusion	Audited conclusion
Generalization	Learned dynamics forecast accurately.	Error rises about fourfold; simple baselines perform better.
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Disclosure	Withholding constants improves forecasts.	Withholding constants increases pooled error by 0.069 rad.
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler-Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models such as Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024] are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We evaluate these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Here the output is a numerical state at a specified time in a chaotic trajectory.

Data-driven identification and abstention. SINDy fits sparse governing equations from measured trajectories [Brunton et al., 2016]; nonlinear vector autoregression provides a compact reservoir-computing formulation [Gauthier et al., 2021]. Structural identifiability asks which parameter combinations can be recovered from observed outputs, including under parameter symmetries [Borgqvist et al., 2024]. Selective prediction ordinarily reports risk together with coverage [Geifman and El-Yaniv, 2017]. Here failed cells remain in the main score because models are compared at fixed task coverage.

3 Benchmark Design

Choice of systems. We use the k -link pendulum to vary the number of coupled links while retaining a high-precision numerical solution. Its constants can also be hidden without changing the underlying trajectory, which permits the matched test in §5.2. The smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system. We did not rerun the LLM or disclosure experiments on Lorenz.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the

Table 2: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except <code>hidden</code>	—

72 RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference
73 by construction, not an independent competitor. Euler and the symplectic integrator are distinct
74 known-physics methods. Inputs available to each family are stated in §3.3.

75 3.2 Regimes and grid

76 Three regimes test generalization to non-standard constants (Table 8). In `changed_hidden` the
77 constants (g, L, m , damping) are not disclosed to the model. This is a matched metadata-disclosure
78 test; a single state contains no observed dynamics from which to identify the constants. An evaluation
79 cell is a tuple (model, k , regime, horizon), with horizons $\{0.01, 1, 10, 60\}$ s. The full LLM grid also
80 varies input modality and prompting; every comparison here uses the coordinate, direct setting, and
81 the other axes appear only in the released records.

82 **Out-of-sample protocol.** Learned dynamics models are evaluated off their training data. We
83 generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions
84 (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed
85 by `movement_id`. The cross-family comparison (§5.1) uses horizons $\{1, 10\}$ s, where chaos has
86 begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model; in-sample error
87 appears only once, to quantify the size of the generalization gap (§5.1).

88 3.3 Inputs and grouping

89 Table 2 states what each family receives. Chronos is a time-series model, so we evaluate it in its
90 standard form: a history-conditioned forecaster with a 10 s pre- $t=0$ context window. The LLMs and
91 learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells
92 are equalized by withholding hidden constants and using normal-regime prior constants. A separate
93 asymmetry runs in the opposite direction: the LLMs are large-scale proprietary models, whereas
94 the learned models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B).
95 The behavior of larger physics-informed networks is outside the scope of this study. We separate
96 known-physics methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned
97 dynamics); the former are numerical references.

98 3.4 Metrics

99 The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over
100 links and trajectories. We also track Cartesian bob error, angular-velocity sign match, and energy
101 drift.

102 **Reliability-adjusted error.** A cell is answered only if the model returns a parseable, finite prediction.
103 Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign
104 unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform
105 guess); we also report the strict upper bound π . Answering is thus part of the score.

106 **Verification record.** For each evaluation cell we retain the trajectory identifier, model and de-
107 ployment name, prompt, raw response, parser outcome, prediction, prompt and completion token
108 counts, and score. The released configuration fixes the data-generation solver, held-out seed, horizons,
109 prompts, timeout, and temperature. Two standards apply here. Local methods are reproducible:
110 their checkpoints and seeds regenerate the predictions exactly. Hosted models are not, because
111 a deployment can change or be withdrawn between evaluations, so for them the record is what

remains. We therefore release the per-cell responses rather than summary scores alone, which keeps the reported API evaluation auditable after the deployments it measured are gone.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

Disclosure gap. We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, we also run a matched experiment (§5.2) that toggles disclosure on identical trajectories.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (hosted deployments, June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing assumptions in the released analysis script.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU in its standard history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels, resampled to up to 512 points, forecasts to the target horizon (prediction length ≤ 64 steps). The comparison uses `chronos-2` in univariate and multivariate modes; the latter forecasts all $2k$ channels jointly. The paired mode comparison uses 20 trajectories per (k, regime) cell.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation and is trained at full scale (67k samples, hidden 256) on a CUDA GPU. The primary models are trained on nine trajectories, one for each (k, regime) cell (Appendix B). A nested budget study trains fresh Neural ODE and HNN checkpoints with $\{1, 5, 20\}$ trajectories per eligible cell and seeds $\{101, 102, 103\}$. The Neural ODE uses all nine cells; because damping violates Hamiltonian conservation, the HNN budget study uses only the three undamped normal-regime cells. Every checkpoint is evaluated on the same two held-out seeds.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 generates the data and is included as a numerical reference, not an independent competitor.

Baselines. Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the learned models.

On the balanced two-seed subset we also fit SINDy with a degree-two polynomial library and STLSQ, and nonlinear vector autoregression (NVAR) with two delays, degree-two features, and ridge regression. Divergent rollouts are retained as failures and receive the same $\pi/2$ penalty.

Lorenz replication. A secondary Lorenz–63 test compares numerical references, persistence, SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on five five-trajectory held-out seeds. We report training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s;

Table 3: Out-of-sample cross-family comparison on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 generates the data and is included as a numerical reference. Configurations first evaluated in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$ (full-scale)	1.356	[1.282, 1.430]	94%	learned

Appendix C gives the full configuration. HNN and LNN are omitted because Lorenz–63 is dissipative and non-Hamiltonian.

5 Results

5.1 Counting failed responses

The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model is evaluated on that grid. The Answered column in Table 3 gives the fraction of cells with valid predictions. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for LLMs they do not include per-call sampling variation.

Table 3 ranks the models on the pendulum under the access conditions in §3.3. The ranking is conditional on those inputs, this training budget, and this system; Section 5.4 tests whether it transfers. The known-equation integrators sit at the top as references (RK4 0.059 at a finer step, symplectic 0.205, Euler 0.405), and every learned dynamics model lands closer to the $\pi/2$ random baseline than to the integrators. The lowest-error black-box entry is the frontier LLM kimi (0.538 rad, 99% answered), and its marginal interval excludes the nearest baseline’s. That is a statement about one recorded June 2026 deployment under these evaluation conditions, not about model families; the Lorenz test rejects the family reading.

After equalizing the hidden-regime inputs, nearest-neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972), though those two intervals overlap. The stable learned models score about 0.26 rad on their training trajectories and 0.97–1.14 rad on the held-out set, so in-sample evaluation understates their error by about a factor of four.

Two further data-driven methods appear only on the balanced two-seed subset, so we report them in Table 4 rather than in the main comparison: sparse identification (SINDy, 0.883 rad on that subset) falls between the Chronos family and the learned models, and nonlinear vector autoregression (NVAR, 1.248) is among the weakest data-driven methods, near constant velocity.

185 **Effect of missing predictions.** grok answers 82% of the grid and lnn 94%. Their ranks fall when
 186 missing cells receive $\pi/2$. An answered-cells-only mean omits the cases on which these models
 187 failed and gives both a lower reported error.

188 5.2 Matched disclosure test

189 On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad,
 190 $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). Read directly, this looks favorable
 191 to withholding constants, but the regimes use different trajectories, and the hidden ones are slower
 192 and less divergent, so the comparison cannot speak to parameter recovery. When we instead toggle
 193 disclosure on identical trajectories (Table 7), the pooled effect reverses: hiding the constants increases
 194 error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with disclosure helping strongly at $k=1$ and only a small
 195 reverse effect at $k=3$. The unadjusted hidden-regime advantage is thus a difficulty confound.

196 The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto$
 197 (am, ad) and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from
 198 an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. Asked to
 199 name the constants from a single state, the LLMs mostly returned Earth-standard defaults, which is
 200 evidence of a prior, not of recovery. The released numerical check verifies both invariances to below
 201 10^{-13} over one-second trajectories.

202 5.3 Replication across seeds and training budgets

203 We reran the local pendulum subset on an independent held-out seed with five trajectories per
 204 (k, regime) cell; the original seed has 20. Table 4 reports each seed separately and gives their
 205 unweighted mean. The selected ordering is unchanged. The numerical references have the lowest
 206 errors, Chronos-2 precedes the learned models, and HNN-rollout has the lowest error among the
 207 selected learned models. Two seeds do not estimate seed-level uncertainty. The LNN needs a note: its
 208 low coverage was a hardware artifact, not model behavior. Its inference solves a Hessian at each step,
 209 which exhausted the 8 GB GPU on the longest rollouts and returned those cells as failures. Re-running
 210 the LNN on CPU is the same evaluation with the same forecasts; it only recovers the dropped cells,
 211 raising primary-seed coverage from 66% to 94%. We use this CPU evaluation for the LNN on both
 212 seeds (the second answers all 90 cells, equal-seed rate 97%), so its numbers are produced identically
 213 and are comparable across seeds. The cells that still fail are genuine autograd-ODE divergences,
 214 scored at $\pi/2$ like any other failure, and the mean stays high at 1.366 rad.

215 **Training-budget check.** With three training initializations and the two held-out seeds, Neural ODE
 216 error decreases from 1.090 at one trajectory per cell to 0.968 at five and 0.861 at twenty (Table 9).
 217 HNN error on the normal regime falls from 1.269 to 0.974 and then remains 0.981. The HNN and
 218 Neural ODE rows cover different regimes and are not an architecture comparison. More data helps
 219 the Neural ODE monotonically, the HNN gains plateau by 15 trajectories, and neither closes the gap
 220 to known-equation integration.

221 5.4 Cross-system test: Lorenz-63

222 We test whether the pendulum conclusions transfer to Lorenz-63, a chaotic system with a polynomial
 223 vector field. They do not. Both the identifiability result and the model ordering change.

224 The identifiability verdict flips. On the pendulum the physical constants are not recoverable from an
 225 angle trajectory (§5.2); on Lorenz, SINDy recovers the governing equations themselves, not merely
 226 their forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$
 227 to coefficient error 2×10^{-13} (Appendix C); forecast error stays below 2×10^{-5} in every seed.
 228 The difference is the setting, not the audit. Lorenz-63 lies inside SINDy’s hypothesis class and is
 229 observed in full state, so its parameters are identifiable, whereas the pendulum’s absolute constants
 230 are not identifiable from angles under any method. Identifiability is a property of the observation
 231 model and the hypothesis class, and the check makes it explicit before recovery is scored.

232 The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25
 233 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and
 234 0.277 ± 0.063 at horizons 0.1, 0.5, and 1.0 s (mean $\pm$ standard error over seeds), below both Chronos-

2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate 0.126 ± 0.027 , 0.700 ± 0.102 , 0.732 ± 0.085). NVAR is close at 0.1 s (0.012) but drifts to 0.429 by 1.0 s, a persistence baseline trails (0.900 at 1 s), and divergent rollouts are scored at $\pi/2$ as on the pendulum (Table 4). This is the reverse of the pendulum, where the trained models lost to trivial baselines: here a system-trained model beats the zero-shot foundation model. As on the pendulum the families see different inputs (SINDy the library, the Neural ODE target-system training, Chronos a history window), so this is not an architecture race; the cross-system result rejects the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable seed. The Neural ODE beats both Chronos-2 modes in every one of the 15 seed-by-horizon comparisons, and SINDy stays below 1.7×10^{-5} in every cell. The identifiability half of the contrast is not a sampling claim at all: the pendulum symmetries are exact and verified to below 10^{-13} , and Lorenz-63 lies in SINDy’s library by construction.

The Lorenz test includes the local methods for which the comparison is well defined. It excludes the API LLM grid and the Hamiltonian and Lagrangian networks; a Hamiltonian model does not match the dissipative Lorenz flow. Those omissions limit what the test can establish, but not what it can refute: the pendulum result is only interesting if it generalizes, and one system on which the ordering and the identifiability verdict both invert is enough to withdraw that generalization. We claim no more than that, and in particular the Lorenz ordering is not offered as the correct one.

5.5 Real data: the reference model becomes the imperfect one

On real hardware the RK4 integrator is no longer ground truth but the imperfect model. We use an open video-tracked double pendulum [Myers et al., 2020] (three free-drop trials, 500 Hz, published point-mass constants, $\sim 0.4^\circ$ median tracking noise); Trial 1 only calibrates (nearest-neighbor database, SINDy fit, one-parameter damping fit), and trials 2–3 supply 120 held-out states scored as before at $\{0.1, 0.5, 1.0\}$ s (Table 14, Appendix F).

RK4 with the published constants leads at 0.1 s (0.084 rad, about $14\times$ the tracking noise) but reaches 1.25 rad by 1.0 s, indistinguishable from persistence (1.23); unmodeled friction and the point-mass simplification consume its advantage within a second. The fitted damping trades horizons rather than dominating (0.89 vs 0.82 at 0.5 s, 1.15 vs 1.25 at 1.0 s). SINDy, exact on Lorenz, sits at baseline level here (1.54); the hardware’s dynamics are not in its library. Chronos-2 answers 89% of cells, since states shortly after the drop lack its 5 s history; failure accounting charges them at $\pi/2$. Every check transfers, and the ordering differs from both synthetic systems.

5.6 Diagnostics

Behavior by horizon. Out-of-sample error by horizon appears in Table 12 (Appendix E). At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed- k Neural ODE is mid-pack at both horizons (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 5 and 6 (Appendix D) report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The changed_hidden regime has the lowest error for almost every model (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19; the previous values favored models given the true constants. The low-gravity hidden constants ($g=1.62$) produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§5.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged. The original bootstrap’s treatment of correlated cells does not determine the reported gaps.

Training-objective check. Table 10 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 11 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged. Changing the integrator therefore does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

6 Discussion

The audit changes five conclusions supported by the raw comparison. Held-out evaluation exposes a roughly fourfold learned-model generalization gap; failure accounting penalizes incomplete coverage; the matched disclosure test reverses the disclosure effect; exact symmetries rule out absolute parameter recovery; and Lorenz-63 changes the model ordering. These are changes to the scientific claim, not to presentation.

More data improves the Neural ODE from 1.090 to 0.861 rad without closing the gap to known-equation integration; HNN improvement plateaus. On Lorenz, SINDy is nearly exact because the vector field lies in its library, while a system-trained Neural ODE beats zero-shot Chronos-2. Performance therefore depends on the system, observations, hypothesis class, data, and access. A scalar leaderboard does not encode those conditions.

The API LLM evaluation is the only billed run. Its per-cell token records give a nominal cost of \$110.38 for 3,960 forecasts, or about \$0.03 each under the June 2026 deployments and pricing. The local methods incur compute cost without a per-query charge.

7 Limitations

The full LLM grid uses one held-out seed; the local replication adds a second with fewer trajectories per cell. The budget study uses three training initializations and two held-out seeds. Inputs differ by method as listed in Table 2. The Lorenz test uses five trajectories in each of five held-out seeds and excludes API LLMs, HNN, and LNN; the hardware check uses three trials of one apparatus with filtered velocities. SINDy’s result depends on a library containing the true polynomial degree. Predictions are single endpoint states, conservation tests cover only the undamped regime, and we cannot rule out pendulum-like trajectories in the foundation models’ training corpora.

8 Conclusion

Correct scores can support wrong scientific conclusions. Here, generalization, failure accounting, matched disclosure, identifiability, and cross-system transfer each change a claim supported by the raw leaderboard, and forecasting benchmarks should report those checks, training budgets, and information access as part of the result. The synthetic systems make each check’s effect visible against exact ground truth; the hardware check runs the same audit where the reference model itself is imperfect, the setting this work was built to reach.

References

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research (TMLR)*; arXiv:2403.07815, 2024.
- Johannes G. Borgqvist, Alexander P. Browning, Fredrik Ohlsson, and Ruth E. Baker. Framing global structural identifiability in terms of parameter symmetries. *arXiv preprint arXiv:2410.03757*, 2024.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15): 3932–3937, 2016. doi: 10.1073/pnas.1517384113.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. *ICLR 2020 Workshop on Integration of Deep Neural Models and Differential Equations*; arXiv:2003.04630, 2020.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2310.10688.
- Daniel J. Gauthier, Erik Bollt, Aaron Griffith, and Wendson A. S. Barbosa. Next generation reservoir computing. *Nature Communications*, 12:5564, 2021. doi: 10.1038/s41467-021-25801-2.
- Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2110.05266.
- Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.07820.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.14858.
- Audun D. Myers, Joshua R. Tempelman, David Petrushenko, and Firas A. Khasawneh. Low-cost double pendulum for high-quality data collection with open-source video tracking and analysis. *HardwareX*, 8: e00138, 2020. doi: 10.1016/j.ohx.2020.e00138. Dataset: Mendeley Data, doi:10.17632/z4hvxjgtbz.2.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02592.
- Xin Xu, Qiyun Xu, Tong Xiao, Tianhao Chen, Yuchen Yan, Jiaxin Zhang, Shizhe Diao, Can Yang, and Yang Wang. Ugphysics: A comprehensive benchmark for undergraduate physics reasoning with large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.00334.
- Yuanzhao Zhang and William Gilpin. Zero-shot forecasting of chaotic systems. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2409.15771.

A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

381 **System prompt (direct / no-CoT).**

382 You are a careful physicist predicting the future state of a k -pendulum (a chain of
383 k point masses on massless rods, hinged in a planar chain, swinging under gravity
384 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
385 no code fences. Schema:
386 {"theta": [...k floats...], "omega": [...k floats...]}
387 theta_i is in radians (link i measured from straight-down,
388 positive counter-clockwise)
389 and omega_i is angular velocity in rad/s.

390 **System prompt (chain-of-thought).** Same preamble, then: "Reason step-by-step about the dynamics,
391 then output your final answer. Wrap the final answer in <answer>...</answer> tags containing ONLY a
392 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}."

393 **User prompt (coordinate modality).**

394 Number of links $k = \{k\}$
395 Physical constants:
396 g (gravity, m/s²): {g}
397 L (rod lengths, m): {L}
398 m (point masses, kg): {m}
399 damping (viscous coefficient on omega): {damping}
400 Initial state at $t=0$:
401 theta_0 (rad): {theta0}
402 omega_0 (rad/s): {omega0}
403 Predict the state at $t = \{T\}$ seconds.
404 Return theta and omega as length- $\{k\}$ JSON arrays.

405 **Hidden-constants regime.** The constants block above is replaced by:

406 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
407 masses, and damping are NOT disclosed. You must infer dynamics from the initial
408 conditions alone.

409 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
410 <answer>...</answer> tags; the first {...} object is JSON-parsed and validated to contain length- k theta
411 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

412 **B Training and compute**

413 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, trained
414 on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
415 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
416 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
417 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
418 trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
419 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
420 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

421 **Reproducibility records.** The anonymized artifact accompanying this submission contains the code,
422 prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison
423 (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep;
424 the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All
425 dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at
426 temperature 0. The reported hosted-model numbers recompute from the saved records; re-calling the API is a
427 new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

428 **C Seed and system replication**

429 Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, one fixed training
430 seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed contains five

trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} . NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	primary seed	second seed	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

436

437 D Breakdowns by link count and regime

438 Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by
439 number of links (Table 5) and by regime (Table 6); models in leaderboard order. Cells a model never produced
440 (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 3.

441 E Additional result tables

442 Full per-model values for the horizon breakdown (§5.6) and the predictability horizon (§5.2), together with the
443 regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity
444 studies. The corresponding summary statistics appear in the main text.

445 F Real-data check details

446 The hardware dataset [Myers et al., 2020] is released under GPL-3 with video tracking of both link angles at
447 500 Hz and per-frame uncertainty estimates (median 0.20° upper link, 0.43° lower). We use the published
448 simplified point-mass parameters ($L = [0.172, 0.143]$ m, $m = [0.311, 0.111]$ kg, $g = 9.81$). Angular velocities
449 come from a Savitzky–Golay filter (window 42 ms, cubic, analytic derivative). The recordings begin with the
450 pendulum positioned by hand, so the free drop is detected as the first time either link exceeds 4 rad/s; evaluation
451 states are drawn evenly from 0.2 to 24 s after the drop, 60 per trial. The tracker zeroes its angle at the first frame,
452 leaving a gravity-referenced offset per link that we select from $\{0, \pi\}$ by which choice lets the physics model
453 track at a 20 ms horizon; the selected offsets are zero for every trial, with 20 ms error 0.005–0.007 rad against

Table 5: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

Table 6: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §5.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 7: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to Table 6.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	-0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			-0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

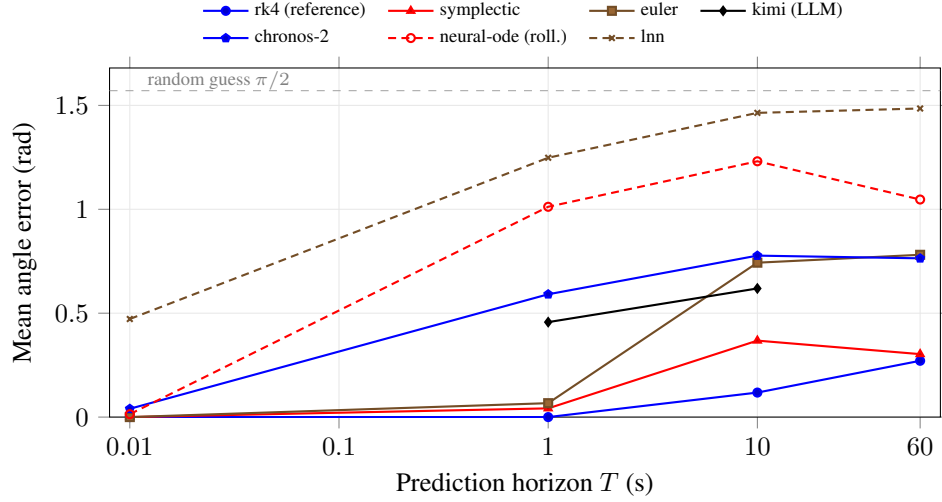


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 8: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 9: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 10: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 11: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 12: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.248	1.464

Table 13: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means no horizon, including the first, stayed below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01

Table 14: Real-data check: mean wrapped angle error (rad) over both links on trials 2–3 (120 held-out states), failures at $\pi/2$. The physics rows use the published constants; “damped” adds the Trial-1-fitted viscous coefficient. Tracking noise floor ≈ 0.006 rad.

Model	0.1 s	0.5 s	1.0 s	Answered
physics-rk4	0.084	0.820	1.248	100%
physics-rk4-damped	0.086	0.892	1.145	100%
nearest-neighbor (cross-trial)	0.371	0.849	1.117	100%
chronos2	0.611	1.160	1.223	89%
constant-velocity	0.345	1.402	1.576	100%
sindy-real	0.490	1.501	1.535	99%
persistence	0.934	1.654	1.230	100%

454 a 0.006 rad tracking floor. The damping grid $\{0, 5 \times 10^{-4}, \dots, 2 \times 10^{-2}\}$ selects 10^{-3} on Trial 1 at the 0.5 s
455 horizon. SINDy uses a degree-two polynomial plus first-order Fourier library on Trial 1 with Savitzky–Golay
456 derivatives; Chronos-2 receives a 5 s context resampled to 50 Hz and forecasts at most 50 steps. Failures
457 (non-finite rollouts, missing context) score $\pi/2$.